

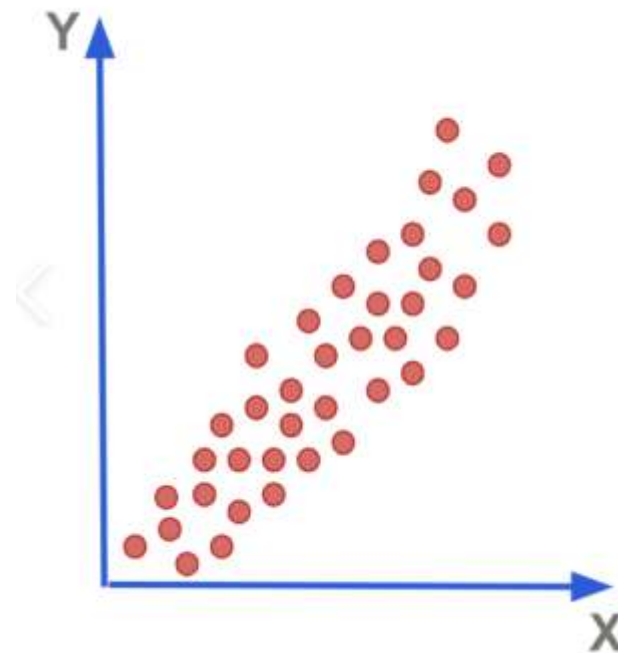
信息学中的概率统计

王若松

前沿计算研究中心
北京大学

1. 回归分析

- ▶ **点估计**：估计分布中所含有的未知参数
- ▶ **回归分析**：估计变量之间的关系
- ▶ 例1：给定同一个电阻不同电流下电压的测量数据，估计电压和电流间的关系
- ▶ 例2：估计广告投入和利润回报间的关系
- ▶ 给定数据 $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ ，估计 y 与 x 的关系
- ▶ **线性相关关系**： $y = \alpha + \beta x + \epsilon$

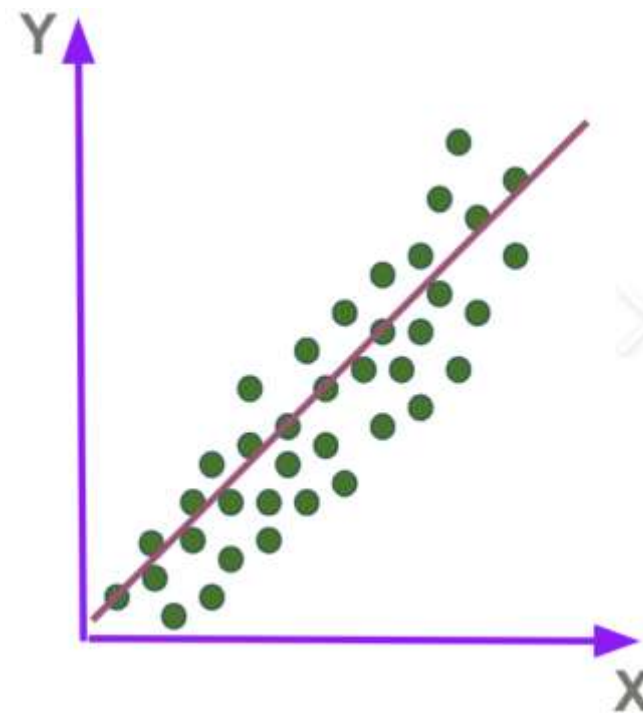


1. 回归分析

- ▶ **回归分析：** 估计变量之间的关系
- ▶ 给定数据 $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, 估计 y 与 x 的关系
- ▶ **线性相关关系：** $y = \alpha + \beta x + \epsilon$
 - ▶ α 与 β 为需要估计的未知参数
 - ▶ ϵ 为误差, $E(\epsilon) = 0$, $\text{Var}(\epsilon) = \sigma^2$, σ^2 为未知参数
 - ▶ x 可以精确测量或严格控制
- ▶ 目标：利用数据 $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, 给出 α 和 β 的估计 $\hat{\alpha}$ 和 $\hat{\beta}$
- ▶ 假设： $y_i = \alpha + \beta x_i + \epsilon_i$
 - ▶ $E(\epsilon_i) = 0$, $\text{Var}(\epsilon_i) = \sigma^2$, σ^2 为未知参数, 且 ϵ_i 相互独立

1. 回归分析

- ▶ 目标：利用数据 $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ ，给出 α 和 β 的估计 $\hat{\alpha}$ 和 $\hat{\beta}$
- ▶ 假设： $y_i = \alpha + \beta x_i + \epsilon_i$ ， $E(\epsilon_i) = 0$ ， $\text{Var}(\epsilon_i) = \sigma^2$ ，且 ϵ_i 相互独立
- ▶ 给定估计 $\hat{\alpha}$ 和 $\hat{\beta}$
- ▶ **经验回归函数**： $\hat{y} = \hat{\alpha} + \hat{\beta}x$
- ▶ 给定 $x = x_0$ ， $\hat{y}_0 = \hat{\alpha} + \hat{\beta}x_0$ 为**预测值或拟合值**



2. 最小二乘估计

- ▶ 目标：利用数据 $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ ，给出 α 和 β 的估计 $\hat{\alpha}$ 和 $\hat{\beta}$
- ▶ 定义： $Q(\alpha, \beta) = \sum_{i=1}^n (y_i - \beta x_i - \alpha)^2$
- ▶ 选择 α 和 β ，最小化 $Q(\alpha, \beta)$ 。称得到的 $\hat{\alpha}$ 和 $\hat{\beta}$ 为**最小二乘估计**
- ▶ 计算问题：给定数据 $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ ，如何计算最小二乘估计
- ▶ 统计性质：最小二乘估计有哪些统计性质
- ▶ 预测：给定新数据 x_0 ，如何估计 y_0

2. 最小二乘估计

- ▶ 目标：利用数据 $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ ，给出 α 与 β 的估计 $\hat{\alpha}$ 和 $\hat{\beta}$
- ▶ 定义： $Q(\alpha, \beta) = \sum_{i=1}^n (y_i - \beta x_i - \alpha)^2$
- ▶ 选择 α 和 β ，最小化 $Q(\alpha, \beta)$ 。称得到的 $\hat{\alpha}$ 和 $\hat{\beta}$ 为**最小二乘估计**

- ▶ **正规方程：**
- ▶ $\frac{\partial Q}{\partial \beta} = -2 \sum x_i \cdot (y_i - \beta x_i - \alpha) = 0$
- ▶ $\frac{\partial Q}{\partial \alpha} = -2 \sum (y_i - \beta x_i - \alpha) = 0$

2. 最小二乘估计

► **正规方程:**

► $\frac{\partial Q}{\partial \beta} = -2\sum x_i \cdot (y_i - \beta x_i - \alpha) = 0$

► $\frac{\partial Q}{\partial \alpha} = -2\sum (y_i - \beta x_i - \alpha) = 0$

► $\frac{\partial Q}{\partial \alpha} = 0 \Rightarrow \alpha = \frac{1}{n}(\sum y_i - \beta \sum x_i)$

► $\frac{\partial Q}{\partial \beta} = 0 \Rightarrow \sum x_i \cdot (y_i - \beta x_i - \alpha) = 0 \Rightarrow \sum x_i y_i - \beta \sum x_i^2 - \frac{1}{n} \sum x_i \cdot (\sum y_i - \beta \sum x_i) = 0$

► $\hat{\beta} = \frac{\sum x_i y_i - \frac{1}{n} \sum x_i \cdot \sum y_i}{\sum x_i^2 - \frac{1}{n} (\sum x_i)^2}, \hat{\alpha} = \frac{1}{n} \sum y_i - \hat{\beta} \cdot \frac{1}{n} \sum x_i$

2. 最小二乘估计

- ▶ 目标：利用数据 $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ ，给出 α 与 β 的估计 $\hat{\alpha}$ 和 $\hat{\beta}$
- ▶ 定义： $Q(\alpha, \beta) = \sum_{i=1}^n (y_i - \beta x_i - \alpha)^2$
- ▶ 选择 α 和 β ，最小化 $Q(\alpha, \beta)$ 。称得到的 $\hat{\alpha}$ 和 $\hat{\beta}$ 为**最小二乘估计**
- ▶ $\hat{\beta} = \frac{\sum x_i y_i - \frac{1}{n} \sum x_i \cdot \sum y_i}{\sum x_i^2 - \frac{1}{n} (\sum x_i)^2}$, $\hat{\alpha} = \frac{1}{n} \sum y_i - \hat{\beta} \cdot \frac{1}{n} \sum x_i$
- ▶ 定义 $\bar{x} = \frac{1}{n} \sum x_i$, $\bar{y} = \frac{1}{n} \sum y_i$
- ▶ $s_{xx} = \sum (x_i - \bar{x})(x_i - \bar{x}) = \sum x_i^2 - n \cdot (\bar{x})^2$
- ▶ $s_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - \sum x_i \cdot \bar{y} - \sum y_i \cdot \bar{x} + n \cdot \bar{x} \cdot \bar{y} = \sum x_i y_i - n \cdot \bar{x} \cdot \bar{y}$
- ▶ $\hat{\beta} = \frac{s_{xy}}{s_{xx}}$, $\hat{\alpha} = \bar{y} - \hat{\beta} \cdot \bar{x}$

2. 最小二乘估计

- ▶ 假设: $y_i = \alpha + \beta x_i + \epsilon_i$, $E(\epsilon_i) = 0$, $\text{Var}(\epsilon_i) = \sigma^2$, 且 ϵ_i 相互独立
- ▶ 注意到 $\sum(x_i - \bar{x}) = 0$
- ▶ $s_{xy} = \sum(x_i - \bar{x})(y_i - \bar{y}) = \sum(x_i - \bar{x})(\alpha + \beta x_i + \epsilon_i - \bar{y}) = \sum(x_i - \bar{x})(\beta x_i + \epsilon_i)$
- ▶
$$\hat{\beta} = \frac{\sum(x_i - \bar{x})(\beta x_i + \epsilon_i)}{s_{xx}} = \sum \epsilon_i \cdot \frac{(x_i - \bar{x})}{s_{xx}} + \frac{\sum(x_i - \bar{x}) \cdot \beta x_i}{s_{xx}}$$
- ▶
$$= \sum \epsilon_i \cdot \frac{(x_i - \bar{x})}{s_{xx}} + \frac{\sum(x_i - \bar{x}) \cdot \beta (x_i - \bar{x})}{s_{xx}} = \sum \epsilon_i \cdot \frac{(x_i - \bar{x})}{s_{xx}} + \beta$$
- ▶
$$\hat{\alpha} = \bar{y} - \hat{\beta} \cdot \bar{x} = \beta \cdot \bar{x} + \alpha + \frac{1}{n} \sum \epsilon_i - \hat{\beta} \cdot \bar{x} = \alpha + \frac{1}{n} \sum \epsilon_i + (\beta - \hat{\beta}) \cdot \bar{x}$$
- ▶
$$= \alpha + \sum \epsilon_i \cdot \left(\frac{1}{n} - \frac{x_i - \bar{x}}{s_{xx}} \cdot \bar{x} \right)$$

2. 最小二乘估计

- ▶ 假设: $y_i = \alpha + \beta x_i + \epsilon_i$, $E(\epsilon_i) = 0$, $\text{Var}(\epsilon_i) = \sigma^2$, 且 ϵ_i 相互独立
- ▶ $\hat{\beta} = \beta + \sum \epsilon_i \cdot \frac{(x_i - \bar{x})}{s_{xx}}$, $\hat{\alpha} = \alpha + \sum \epsilon_i \cdot \left(\frac{1}{n} - \frac{(x_i - \bar{x})}{s_{xx}} \cdot \bar{x} \right)$
- ▶ $E(\hat{\beta}) = \beta, E(\hat{\alpha}) = \alpha$
- ▶ $s_{xx} = \sum (x_i - \bar{x})(x_i - \bar{x})$
- ▶ $\text{MSE}(\hat{\beta}) = \text{Var}(\hat{\beta}) = \sum \text{Var}(\epsilon_i) \cdot \left(\frac{(x_i - \bar{x})}{s_{xx}} \right)^2 = \frac{\sigma^2}{s_{xx}}$
- ▶ $\text{MSE}(\hat{\alpha}) = \text{Var}(\hat{\alpha}) = \sigma^2 \cdot \sum \left(\frac{1}{n} - \frac{(x_i - \bar{x})}{s_{xx}} \cdot \bar{x} \right)^2 = \sigma^2 \sum \left(\frac{1}{n^2} - \frac{2}{n} \cdot \frac{(x_i - \bar{x})}{s_{xx}} \cdot \bar{x} + \left(\frac{(x_i - \bar{x})}{s_{xx}} \cdot \bar{x} \right)^2 \right)$
- ▶ $= \sigma^2 \left(\frac{1}{n} + \frac{(\bar{x})^2}{s_{xx}} \right)$

2. 最小二乘估计

► 假设: $y_i = \alpha + \beta x_i + \epsilon_i$, $E(\epsilon_i) = 0$, $\text{Var}(\epsilon_i) = \sigma^2$, 且 ϵ_i 相互独立

► $\hat{\beta} = \sum \epsilon_i \cdot \frac{(x_i - \bar{x})}{s_{xx}} + \beta$, $\hat{\alpha} = \alpha + \sum \epsilon_i \cdot \left(\frac{1}{n} - \frac{(x_i - \bar{x})}{s_{xx}} \cdot \bar{x} \right)$

► $\text{Var}(\hat{\beta}) = \frac{\sigma^2}{s_{xx}}$, $\text{Var}(\hat{\alpha}) = \sigma^2 \left(\frac{1}{n} + \frac{(\bar{x})^2}{s_{xx}} \right)$

► $\text{Cov}(\hat{\alpha}, \hat{\beta}) = E \left(\sum_i \epsilon_i \cdot \left(\frac{1}{n} - \frac{(x_i - \bar{x})}{s_{xx}} \cdot \bar{x} \right) \cdot \sum_j \epsilon_j \cdot \frac{(x_j - \bar{x})}{s_{xx}} \right)$

► $= E \left(\sum_i \epsilon_i^2 \cdot \left(\frac{1}{n} - \frac{(x_i - \bar{x})}{s_{xx}} \cdot \bar{x} \right) \cdot \frac{(x_i - \bar{x})}{s_{xx}} \right)$

► $= E \left(\sum_i \epsilon_i^2 \cdot \frac{1}{n} \cdot \frac{(x_i - \bar{x})}{s_{xx}} \right) - E \left(\sum_i \epsilon_i^2 \cdot \bar{x} \cdot \left(\frac{(x_i - \bar{x})}{s_{xx}} \right)^2 \right) = -\sigma^2 \cdot \frac{\bar{x}}{s_{xx}}$

2. 最小二乘估计

- ▶ 假设: $y_i = \alpha + \beta x_i + \epsilon_i$, $E(\epsilon_i) = 0$, $\text{Var}(\epsilon_i) = \sigma^2$, 且 ϵ_i 相互独立
- ▶ $\hat{\beta} = \sum \epsilon_i \cdot \frac{(x_i - \bar{x})}{s_{xx}} + \beta$, $\hat{\alpha} = \alpha + \sum \epsilon_i \cdot \left(\frac{1}{n} - \frac{(x_i - \bar{x})}{s_{xx}} \cdot \bar{x} \right)$
- ▶ $\text{Var}(\hat{\beta}) = \frac{\sigma^2}{s_{xx}}$, $\text{Var}(\hat{\alpha}) = \sigma^2 \left(\frac{1}{n} + \frac{(\bar{x})^2}{s_{xx}} \right)$, $\text{Cov}(\hat{\alpha}, \hat{\beta}) = -\sigma^2 \cdot \frac{\bar{x}}{s_{xx}}$
- ▶ 令 $\hat{y}_i = \hat{\alpha} + \hat{\beta} x_i$ 为 x_i 的预测值, $E(\hat{y}_i) = \alpha + \beta x_i \Rightarrow E(\hat{y}_i - y_i) = 0$
- ▶ $\text{Var}(\hat{y}_i) = \text{Var}(\hat{\alpha} + \hat{\beta} x_i) = \text{Var}(\hat{\alpha}) + x_i^2 \cdot \text{Var}(\hat{\beta}) + 2x_i \cdot \text{Cov}(\hat{\alpha}, \hat{\beta})$
- ▶ $= \sigma^2 \left(\frac{1}{n} + \frac{(\bar{x})^2}{s_{xx}} \right) + x_i^2 \cdot \frac{\sigma^2}{s_{xx}} - 2x_i \cdot \sigma^2 \cdot \frac{\bar{x}}{s_{xx}}$
- ▶ $= \sigma^2 \left(\frac{1}{n} + \frac{(\bar{x} - x_i)^2}{s_{xx}} \right)$

2. 最小二乘估计

- ▶ 假设: $y_i = \alpha + \beta x_i + \epsilon_i$, $E(\epsilon_i) = 0$, $\text{Var}(\epsilon_i) = \sigma^2$, 且 ϵ_i 相互独立
- ▶ $\hat{\beta} = \sum \epsilon_i \cdot \frac{(x_i - \bar{x})}{s_{xx}} + \beta$, $\hat{\alpha} = \alpha + \sum \epsilon_i \cdot \left(\frac{1}{n} - \frac{(x_i - \bar{x})}{s_{xx}} \cdot \bar{x} \right)$
- ▶ $\text{Var}(\hat{\beta}) = \frac{\sigma^2}{s_{xx}}$, $\text{Var}(\hat{\alpha}) = \sigma^2 \left(\frac{1}{n} + \frac{(\bar{x})^2}{s_{xx}} \right)$, $\text{Cov}(\hat{\alpha}, \hat{\beta}) = -\sigma^2 \cdot \frac{\bar{x}}{s_{xx}}$
- ▶ 令 $\hat{y}_i = \hat{\alpha} + \hat{\beta} x_i$ 为 x_i 的预测值, $E(\hat{y}_i) = \alpha + \beta x_i$, $\text{Var}(\hat{y}_i) = \sigma^2 \left(\frac{1}{n} + \frac{(\bar{x} - x_i)^2}{s_{xx}} \right)$
- ▶ $\text{Var}(\hat{y}_i - y_i) = \text{Var}(\hat{y}_i) + \text{Var}(y_i) - 2\text{Cov}(\hat{y}_i, y_i)$
- ▶ $= \text{Var}(\hat{y}_i) + \sigma^2 - 2\text{Cov}(\hat{\alpha} + \hat{\beta} x_i, \epsilon_i)$
- ▶ $= \sigma^2 \left(\frac{1}{n} + \frac{(\bar{x} - x_i)^2}{s_{xx}} \right) + \sigma^2 - 2\sigma^2 \left(\frac{1}{n} - \frac{(x_i - \bar{x})}{s_{xx}} \cdot \bar{x} \right) - 2\sigma^2 \frac{(x_i - \bar{x})}{s_{xx}} \cdot x_i$
- ▶ $= \frac{n-1}{n} \sigma^2 + \frac{(\bar{x} - x_i)^2}{s_{xx}} \sigma^2 - 2\sigma^2 \frac{(x_i - \bar{x})}{s_{xx}} \cdot (x_i - \bar{x}) = \frac{n-1}{n} \sigma^2 - \frac{(\bar{x} - x_i)^2}{s_{xx}} \sigma^2$

2. 最小二乘估计

- ▶ 假设: $y_i = \alpha + \beta x_i + \epsilon_i$, $E(\epsilon_i) = 0$, $\text{Var}(\epsilon_i) = \sigma^2$, 且 ϵ_i 相互独立
- ▶ 令 $\hat{y}_i = \hat{\alpha} + \hat{\beta}x_i$ 为 x_i 的预测值, $E(\hat{y}_i) = \alpha + \beta x_i \Rightarrow E(\hat{y}_i - y_i) = 0$
- ▶ $\text{Var}(\hat{y}_i - y_i) = \frac{n-1}{n} \sigma^2 - \frac{(\bar{x}-x_i)^2}{s_{xx}} \sigma^2$
- ▶ $E(\sum(\hat{y}_i - y_i)^2) = \sum \text{Var}(\hat{y}_i - y_i) = (n-1)\sigma^2 - \frac{\sum(\bar{x}-x_i)^2}{s_{xx}} \sigma^2 = (n-2)\sigma^2$
- ▶ $E\left(\frac{1}{n-2} \sum(\hat{y}_i - y_i)^2\right) = \sigma^2$, 也即 $s^2 = \frac{1}{n-2} \sum(\hat{y}_i - y_i)^2$ 为 σ^2 的无偏估计量

2. 最小二乘估计

- ▶ 假设: $y_i = \alpha + \beta x_i + \epsilon_i$, $E(\epsilon_i) = 0$, $\text{Var}(\epsilon_i) = \sigma^2$, 且 ϵ_i 相互独立
- ▶ 更强的假设: $y_i = \alpha + \beta x_i + \epsilon_i$, $\epsilon_i \sim N(0, \sigma^2)$, 且 ϵ_i 相互独立

- ▶ 在新假设下

- ▶ α, β, σ^2 的最大似然估计?
- ▶ 给定新数据 x_0 , 给出 y_0 的置信区间?

- ▶ 似然函数
$$L(\alpha, \beta, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi} \cdot \sigma} e^{-\frac{(y_i - \alpha - \beta x_i)^2}{2\sigma^2}}$$

- ▶ 对数似然函数
$$\ln L(\alpha, \beta, \sigma^2) = \sum_{i=1}^n -\frac{(y_i - \alpha - \beta x_i)^2}{2\sigma^2} - n \cdot \frac{\ln 2\pi}{2} - n \cdot \frac{\ln \sigma^2}{2}$$

2. 最小二乘估计

- ▶ 假设: $y_i = \alpha + \beta x_i + \epsilon_i$, $\epsilon_i \sim N(0, \sigma^2)$, 且 ϵ_i 相互独立
- ▶ 对数似然函数 $\ln L(\alpha, \beta, \sigma^2) = \sum_{i=1}^n -\frac{(y_i - \alpha - \beta x_i)^2}{2\sigma^2} - n \cdot \frac{\ln(2\pi)}{2} - n \cdot \frac{\ln \sigma^2}{2}$
- ▶ 对于固定的 σ^2 , 最大化似然函数等价于最大化 $-\sum_{i=1}^n (y_i - \alpha - \beta x_i)^2$
- ▶ 等价于最小二乘估计 $\hat{\alpha}$ 和 $\hat{\beta}$
- ▶ σ^2 的最大似然估计?
- ▶ $\frac{\partial L}{\partial \sigma^2} = \frac{(y_i - \alpha - \beta x_i)^2}{2(\sigma^2)^2} - \frac{n}{2\sigma^2} = 0$
- ▶ $\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{\alpha} - \hat{\beta} x_i)^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$
- ▶ $\hat{\sigma}_{\text{MLE}}^2$ 为有偏估计

2. 最小二乘估计

- ▶ 假设: $y_i = \alpha + \beta x_i + \epsilon_i$, $\epsilon_i \sim N(0, \sigma^2)$, 且 ϵ_i 相互独立
- ▶ 最小二乘估计 $\hat{\alpha}$ 和 $\hat{\beta}$ 服从何种分布?

- ▶ 回顾:

$$\blacktriangleright \hat{\beta} = \sum \epsilon_i \cdot \frac{(x_i - \bar{x})}{s_{xx}} + \beta, \hat{\alpha} = \alpha + \sum \epsilon_i \cdot \left(\frac{1}{n} - \frac{(x_i - \bar{x})}{s_{xx}} \cdot \bar{x} \right)$$

$$\blacktriangleright \text{Var}(\hat{\beta}) = \frac{\sigma^2}{s_{xx}}, \text{Var}(\hat{\alpha}) = \sigma^2 \left(\frac{1}{n} + \frac{(\bar{x})^2}{s_{xx}} \right), \text{Cov}(\hat{\alpha}, \hat{\beta}) = -\sigma^2 \cdot \frac{\bar{x}}{s_{xx}}$$

- ▶ 若 $\epsilon_i \sim N(0, \sigma^2)$, 且 ϵ_i 相互独立, 则 $\hat{\alpha}$ 和 $\hat{\beta}$ 服从二维高斯分布
- ▶ 作业: 令 $s^2 = \frac{1}{n-2} \sum (\hat{y}_i - y_i)^2$, 则有 $\frac{(n-2)s^2}{\sigma^2} \sim \chi^2(n-2)$, 且 s^2 与 $\hat{\alpha}$ 和 $\hat{\beta}$ 独立

2. 最小二乘估计

- ▶ 假设: $y_i = \alpha + \beta x_i + \epsilon_i$, $\epsilon_i \sim N(0, \sigma^2)$, 且 ϵ_i 相互独立
- ▶ 给定新数据 x_0 , 给出 $E(y_0) = \alpha + \beta x_0$ 的估计
- ▶ 点估计: $\hat{\alpha} + \hat{\beta}x_0$
 - ▶ $E(\hat{\alpha} + \hat{\beta}x_0) = \alpha + \beta x_0, \text{Var}(\hat{\alpha} + \hat{\beta}x_0) = \sigma^2 \left(\frac{1}{n} + \frac{(\bar{x} - x_0)^2}{S_{xx}} \right)$
 - ▶ $\hat{\alpha} + \hat{\beta}x_0 \sim N \left(\alpha + \beta x_0, \sigma^2 \left(\frac{1}{n} + \frac{(\bar{x} - x_0)^2}{S_{xx}} \right) \right)$
 - ▶ $\hat{\alpha} + \hat{\beta}x_0 - y_0 \sim N \left(0, \sigma^2 \left(\frac{1}{n} + \frac{(\bar{x} - x_0)^2}{S_{xx}} + 1 \right) \right)$

3. 多元线性回归

- ▶ 多元回归问题
- ▶ 例：估计广告投入 x_1 、产品定价 x_2 以及竞争对手活动 x_3 和利润回报 y 的关系
- ▶ **(多元) 线性相关关系：** $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \epsilon$
 - ▶ ϵ 为误差, $E(\epsilon) = 0$, $\text{Var}(\epsilon) = \sigma^2$, σ^2 为未知参数
 - ▶ $\beta_0, \beta_1, \dots, \beta_k$ 为需要估计的未知参数
- ▶ 目标：利用数据 $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, 给出 $\beta_0, \beta_1, \dots, \beta_k$ 的估计 $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$
- ▶ 假设： $y_i = \beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \cdots + \beta_k x_{i,k} + \epsilon_i$
 - ▶ $E(\epsilon_i) = 0$, $\text{Var}(\epsilon_i) = \sigma^2$, σ^2 为未知参数, 且 ϵ_i 相互独立

3. 多元线性回归

- ▶ 目标：利用数据 $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_n, y_n)$ ，给出 $\beta_0, \beta_1, \dots, \beta_k$ 的估计 $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$
- ▶ 假设： $y_i = \beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \dots + \beta_k x_{i,k} + \epsilon_i$
 - ▶ $E(\epsilon_i) = 0$, $\text{Var}(\epsilon_i) = \sigma^2$, σ^2 为未知参数，且 ϵ_i 相互独立
- ▶ $y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \epsilon_i$
- ▶ $\mathbf{x}_i = [1, x_{i,1}, x_{i,2}, \dots, x_{i,k}] \in \mathbb{R}^{k+1}$
- ▶ $\boldsymbol{\beta} = [\beta_0, \beta_1, \dots, \beta_k]$
- ▶ 最小二乘估计
 - ▶ 定义 $Q(\boldsymbol{\beta}) = \sum_{i=1}^n (y_i - \boldsymbol{\beta}^\top \mathbf{x}_i)^2$
 - ▶ 选择 $\boldsymbol{\beta}$ ，最小化 $Q(\boldsymbol{\beta})$ 。得到的 $\hat{\boldsymbol{\beta}}$ 为**最小二乘估计**
- ▶ 经验回归函数： $\hat{y} = \mathbf{x}^\top \hat{\boldsymbol{\beta}}$

3. 多元线性回归

- ▶ $y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \epsilon_i$
- ▶ $\mathbf{x}_i = [1, x_{i,1}, x_{i,2}, \dots, x_{i,k}] \in \mathbb{R}^{k+1}$
- ▶ $\boldsymbol{\beta} = [\beta_0, \beta_1, \dots, \beta_k]$
- ▶ 最小二乘估计
 - ▶ 定义 $Q(\boldsymbol{\beta}) = \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2$
 - ▶ 选择 $\boldsymbol{\beta}$, 最小化 $Q(\boldsymbol{\beta})$ 。称得到的 $\hat{\boldsymbol{\beta}}$ 为最小二乘估计
- ▶ 矩阵形式: $X \in \mathbb{R}^{n \times (k+1)}$, X 的第 i 行为 $\mathbf{x}_i$, $\mathbf{y} \in \mathbb{R}^n$, $\boldsymbol{\epsilon} \in \mathbb{R}^n$
- ▶ $Q(\boldsymbol{\beta}) = \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2 = \|\mathbf{y} - X\boldsymbol{\beta}\|^2$
- ▶ $\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon}$
- ▶ $E(Y) = X\boldsymbol{\beta}$, $\text{Cov}(Y) = \sigma^2 \cdot I_n$

3. 多元线性回归

- ▶ 最小二乘估计

- ▶ 定义 $Q(\boldsymbol{\beta}) = \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2$
 - ▶ 选择 $\boldsymbol{\beta}$, 最小化 $Q(\boldsymbol{\beta})$ 。称得到的 $\hat{\boldsymbol{\beta}}$ 为最小二乘估计

- ▶ 矩阵形式: $X \in \mathbb{R}^{n \times (k+1)}$, X 的第 i 行为 $\mathbf{x}_i$, $\mathbf{y} \in \mathbb{R}^n$, $\boldsymbol{\epsilon} \in \mathbb{R}^n$

- ▶ $Q(\boldsymbol{\beta}) = \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2 = \|\mathbf{y} - X\boldsymbol{\beta}\|^2$

- ▶ **正规方程:**

- ▶ $\nabla Q = -2X^T(\mathbf{y} - X\boldsymbol{\beta}) = 2X^T X\boldsymbol{\beta} - 2X^T \mathbf{y} = 0$

- ▶ 若 X 列满秩, $\hat{\boldsymbol{\beta}} = (X^T X)^{-1} X^T \mathbf{y}$

- ▶ X 列满秩在一元线性回归中对应条件? 若 X 不列满秩会发生什么?

3. 多元线性回归

- ▶ $y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \epsilon_i$
- ▶ $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$
- ▶ $\mathbf{x}_i = [1, x_{i,1}, x_{i,2}, \dots, x_{i,k}] \in \mathbb{R}^{k+1}$
- ▶ $\boldsymbol{\beta} = [\beta_0, \beta_1, \dots, \beta_k]$

- ▶ **正规方程:**
- ▶ $\nabla Q = -2\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = 2\mathbf{X}^\top \mathbf{X}\boldsymbol{\beta} - 2\mathbf{X}^\top \mathbf{y} = 0$
- ▶ 若 $\mathbf{X}$ 列满秩, $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$

- ▶ $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}) = \boldsymbol{\beta} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\epsilon}$
- ▶ $E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$, $\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$
- ▶ 与一元情况的联系?

3. 多元线性回归

- ▶ $\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}$
- ▶ $\mathbf{X}^T \mathbf{X} = \begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix}$
- ▶ $\det(\mathbf{X}^T \mathbf{X}) = n \sum x_i^2 - (\sum x_i)^2 = n \cdot s_{xx}$
- ▶ 回顾: $s_{xx} = \sum (x_i - \bar{x})(x_i - \bar{x}) = \sum x_i^2 - n \cdot (\bar{x})^2$
- ▶ $(\mathbf{X}^T \mathbf{X})^{-1} = \frac{1}{n \cdot s_{xx}} \begin{bmatrix} \sum x_i^2 & -n\bar{x} \\ -n\bar{x} & n \end{bmatrix}$
- ▶ $\text{Cov}(\hat{\boldsymbol{\beta}}) = \frac{\sigma^2}{n \cdot s_{xx}} \begin{bmatrix} \sum x_i^2 & -n\bar{x} \\ -n\bar{x} & n \end{bmatrix}$
- ▶ 回顾: $\text{Var}(\hat{\beta}) = \frac{\sigma^2}{s_{xx}}, \text{Var}(\hat{\alpha}) = \sigma^2 \left(\frac{1}{n} + \frac{(\bar{x})^2}{s_{xx}} \right), \text{Cov}(\hat{\alpha}, \hat{\beta}) = -\sigma^2 \cdot \frac{\bar{x}}{s_{xx}}$

3. 多元线性回归

- ▶ $y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \epsilon_i$
- ▶ $\mathbf{x}_i = [1, x_{i,1}, x_{i,2}, \dots, x_{i,k}] \in \mathbb{R}^{k+1}$
- ▶ $\boldsymbol{\beta} = [\beta_0, \beta_1, \dots, \beta_k]$

- ▶ **正规方程:**
- ▶ $\nabla Q = -2\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = 2\mathbf{X}^\top \mathbf{X}\boldsymbol{\beta} - 2\mathbf{X}^\top \mathbf{y} = 0$
- ▶ 若 $\mathbf{X}$ 列满秩, $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$

- ▶ $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$
- ▶ 令 $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$
- ▶ $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$

3. 多元线性回归

- ▶ $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$
- ▶ $\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T, \hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$
- ▶ 性质:
- ▶ $\text{tr}(\mathbf{H}) = k + 1, \mathbf{H}^2 = \mathbf{H}, (\mathbf{I} - \mathbf{H})^2 = \mathbf{I} - \mathbf{H}$
- ▶ $\mathbf{H}$ 对称+半正定, $\mathbf{H}\mathbf{X} = \mathbf{X}$
- ▶ $\mathbf{H}$ 为到 $\mathbf{X}$ 列空间上的正交投影
- ▶ $\mathbf{y} - \hat{\mathbf{y}}$ 垂直于 $\mathbf{X}$ 的列空间

3. 多元线性回归

- ▶ $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$
- ▶ $\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T, \hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$
- ▶ 性质:
- ▶ $\text{tr}(\mathbf{H}) = k + 1, \mathbf{H}^2 = \mathbf{H}, (\mathbf{I} - \mathbf{H})^2 = \mathbf{I} - \mathbf{H}$
- ▶ $\mathbf{H}$ 对称+半正定, $\mathbf{H}\mathbf{X} = \mathbf{X}$
- ▶ $\hat{\mathbf{y}} - \mathbf{y} = (\mathbf{H} - \mathbf{I})\mathbf{y} = (\mathbf{H} - \mathbf{I})(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}) = (\mathbf{H} - \mathbf{I})\boldsymbol{\epsilon}$
- ▶ $\text{Cov}(\hat{\mathbf{y}} - \mathbf{y}) = \sigma^2(\mathbf{H} - \mathbf{I})(\mathbf{H} - \mathbf{I})^T = \sigma^2(\mathbf{I} - \mathbf{H})$
- ▶ $E(|\hat{\mathbf{y}} - \mathbf{y}|^2) = \sigma^2 \text{tr}(\mathbf{I} - \mathbf{H}) = \sigma^2(n - k - 1)$
- ▶ $\frac{1}{n-k-1} |\hat{\mathbf{y}} - \mathbf{y}|^2$ 为 σ^2 的无偏估计量

3. 多元线性回归

- ▶ 给定新数据 $\mathbf{x}_0$, 给出 $E(y_0) = \mathbf{x}_0^\top \boldsymbol{\beta}$ 的估计
- ▶ 点估计: $\mathbf{x}_0^\top \hat{\boldsymbol{\beta}}$
- ▶ $E(\mathbf{x}_0^\top \hat{\boldsymbol{\beta}}) = \mathbf{x}_0^\top \boldsymbol{\beta}$
- ▶ 回顾: $\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$
- ▶ $\text{Var}(\mathbf{x}_0^\top \hat{\boldsymbol{\beta}}) = \sigma^2 \cdot \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0$
- ▶ 几何视角
- ▶ $\mathbf{X}^\top \mathbf{X}$ 刻画了由已有数据描述的椭球
- ▶ $\mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0$ 刻画了新数据 $\mathbf{x}_0$ 与已有数据的“不匹配程度”

3. 多元线性回归

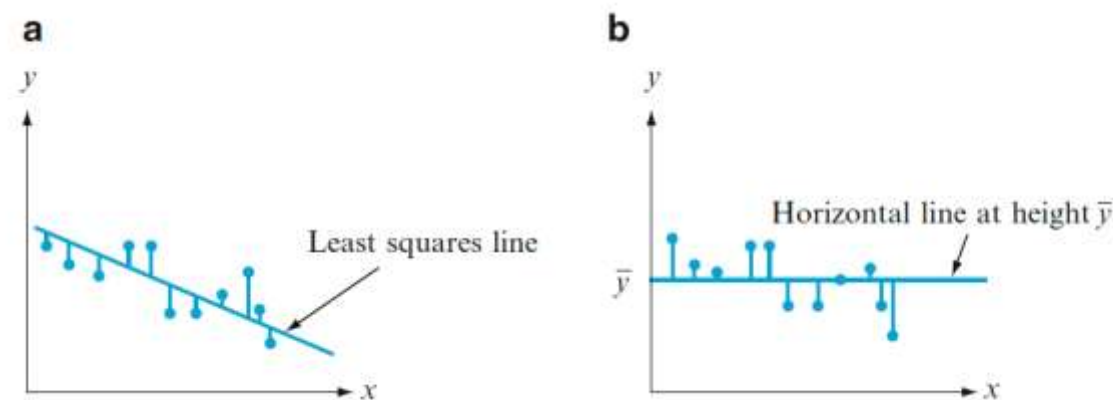
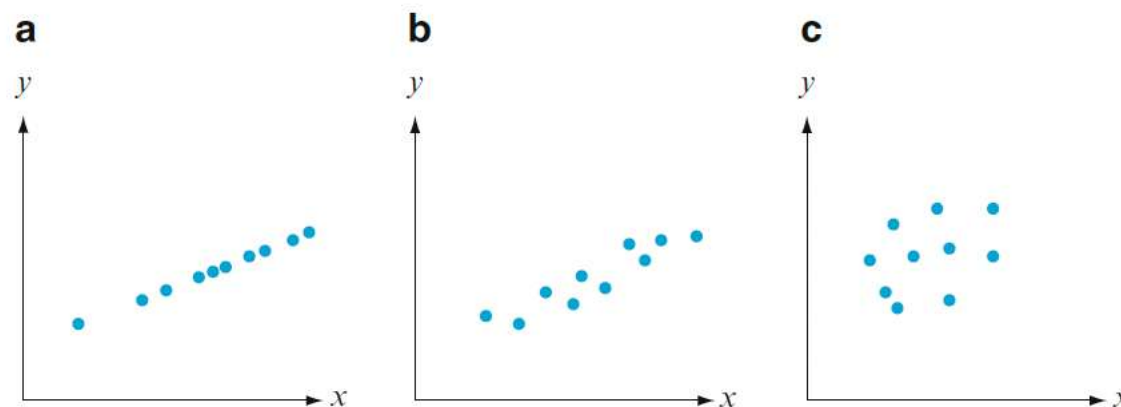
- ▶ 假设: $y_i = \mathbf{x}_i^T \boldsymbol{\beta} + \epsilon_i$, $\epsilon_i \sim N(0, \sigma^2)$, 且 ϵ_i 相互独立
- ▶ 对数似然函数 $\ln L(\boldsymbol{\beta}, \sigma^2) = \sum_{i=1}^n -\frac{(y_i - \boldsymbol{\beta}^T \mathbf{x}_i)^2}{2\sigma^2} - n \cdot \frac{\ln(2\pi)}{2} - n \cdot \frac{\ln \sigma^2}{2}$
- ▶ 对于固定的 σ^2 , 最大化似然函数等价于最大化 $-Q(\boldsymbol{\beta}) = -|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}|^2$
- ▶ 等价于最小二乘估计 $\hat{\boldsymbol{\beta}}$
- ▶ σ^2 的最大似然估计?
- ▶ $\frac{\partial L}{\partial \sigma^2} = \frac{(y_i - \boldsymbol{\beta}^T \mathbf{x}_i)^2}{2(\sigma^2)^2} - \frac{n}{2\sigma^2} = 0$
- ▶ $\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{\boldsymbol{\beta}}^T \mathbf{x}_i)^2 = \frac{1}{n} \cdot |\hat{\mathbf{y}} - \mathbf{y}|^2$
- ▶ $\hat{\sigma}_{\text{MLE}}^2$ 为有偏估计量

3. 多元线性回归

- ▶ 假设: $y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \epsilon_i$, $\epsilon_i \sim N(0, \sigma^2)$, 且 ϵ_i 相互独立
- ▶ $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\mathbf{X} \boldsymbol{\beta} + \boldsymbol{\epsilon}) = \boldsymbol{\beta} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\epsilon}$
- ▶ 回顾: $E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$, $\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$
- ▶ $\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1})$
- ▶ $|\hat{\mathbf{y}} - \mathbf{y}|^2 = |(\mathbf{I} - \mathbf{H})\mathbf{y}|^2 = |(\mathbf{I} - \mathbf{H})(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon})|^2 = |(\mathbf{I} - \mathbf{H})\boldsymbol{\epsilon}|^2$
- ▶ $\mathbf{I} - \mathbf{H}$ 为对称矩阵, 特征值为0或1, 因此 $\mathbf{I} - \mathbf{H} = \mathbf{Q}^\top \boldsymbol{\Lambda} \mathbf{Q}$,
- ▶ $\boldsymbol{\Lambda}$ 为对角矩阵, 前 $n - k - 1$ 个元素为1, 其余元素为0
- ▶ 旋转不变性: $\frac{(\mathbf{I} - \mathbf{H})\boldsymbol{\epsilon}}{\sigma}$ 为 $n - k - 1$ 个独立同分布服从标准正态分布随机变量之和
- ▶ $\frac{|\hat{\mathbf{y}} - \mathbf{y}|^2}{\sigma^2} \sim \chi^2(n - k - 1)$ 。令 $s^2 = \frac{1}{n - k - 1} |\hat{\mathbf{y}} - \mathbf{y}|^2$, 则有 $\frac{(n - k - 1)s^2}{\sigma^2} \sim \chi^2(n - k - 1)$
- ▶ 可证明 s^2 与 $\hat{\boldsymbol{\beta}}$ 相互独立

3. 多元线性回归

- ▶ **确定系数** $r^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$
- ▶ 常用于评估模型的有效性
- ▶ $SST = \sum(y_i - \bar{y})^2$: 总平方和
- ▶ $SSE = \sum(y_i - \hat{y}_i)^2$: 残差平方和
 - ▶ $y_i - \hat{y}_i$: 残差
- ▶ $SSR = \sum(\hat{y}_i - \bar{y})^2$: 回归平方和
- ▶ 作业: $SST = SSE + SSR$
- ▶ $r^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST}$, 有 $0 \leq r^2 \leq 1$
- ▶ r^2 描述了总平方和中, 回归平方和所占的比例, 也即可被模型解释的比例



4. 多元线性回归扩展

- ▶ 最小二乘估计

- ▶ 定义 $Q(\boldsymbol{\beta}) = \sum_{i=1}^n (y_i - \mathbf{x}_i^T \boldsymbol{\beta})^2$
 - ▶ 选择 $\boldsymbol{\beta}$, 最小化 $Q(\boldsymbol{\beta})$ 。称得到的 $\hat{\boldsymbol{\beta}}$ 为最小二乘估计

- ▶ 若 X 列满秩, $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$

- ▶ 若 X 列不满秩, 最优解不唯一

- ▶ X 近似满秩?

- ▶ 回顾: $\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}$

4. 多元线性回归扩展

▶ 岭回归

▶ $Q_\lambda(\boldsymbol{\beta}) = \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2 + \lambda \sum \beta_i^2 = \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2 + \lambda |\boldsymbol{\beta}|^2$

▶ 选择 $\boldsymbol{\beta}$, 最小化 $Q_\lambda(\boldsymbol{\beta})$

▶ $\nabla Q = -2\mathbf{X}^T(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + 2\lambda\boldsymbol{\beta} = 2(\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})\boldsymbol{\beta} - 2\mathbf{X}^T\mathbf{y} = 0$

▶ $\hat{\boldsymbol{\beta}}_\lambda = (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T\mathbf{y}$

▶ $\hat{\boldsymbol{\beta}}_\lambda = (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon})$

▶ $\hat{\boldsymbol{\beta}}_\lambda$ 是否为无偏估计量? 均方误差?

4. 多元线性回归扩展

► 岭回归

► $Q_\lambda(\mathbf{I})$

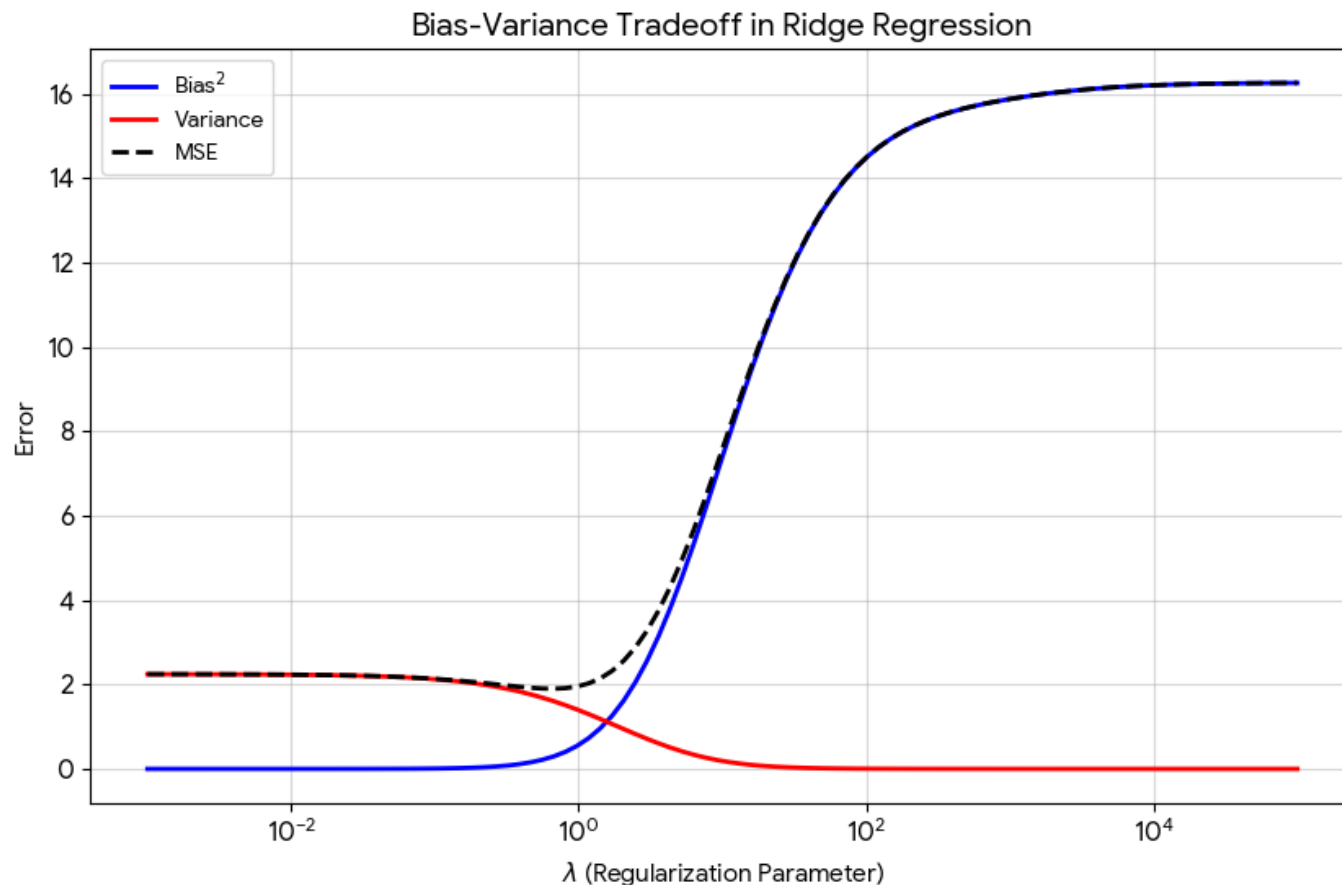
► 选择

► $\hat{\beta}_\lambda = (X^T X + \lambda I)^{-1} X^T \mathbf{y}$

► $\hat{\beta}_\lambda \rightarrow \beta$

► $E(\hat{\beta}_\lambda) = \beta$

► $\text{Cov}(\hat{\beta}_\lambda) = \sigma^2 (X^T X + \lambda I)^{-1}$



$$(X^T X + \lambda I)^{-1} X^T \epsilon$$

► $E(|\hat{\beta}_\lambda - \beta|^2) = \lambda^2 \beta^T (X^T X + \lambda I)^{-2} \beta + \sigma^2 \cdot \text{tr}((X^T X + \lambda I)^{-1} X^T X (X^T X + \lambda I)^{-1})$

3. 多元线性回归

- ▶ 给定 $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, $x_i \in \mathbb{R}^{k+1}$, $y_i \in \mathbb{R}$
- ▶ 令矩阵 $X \in \mathbb{R}^{n \times (k+1)}$, X 的第 i 行为 x_i , $y \in \mathbb{R}^n$
- ▶ X 中某些列可能是多余的, 如何进行选择?

- ▶ 方案1: 枚举列的全部子集, 选择效果最好的子集
 - ▶ 通常使用确定系数 r^2 作为评估标准

- ▶ 方案2: LASSO: $Q(\beta) = \sum_{i=1}^n (y_i - \beta^T x_i)^2 + \lambda \cdot \sum |\beta_i|$
 - ▶ λ 为超参数, 控制选取列数的数量